HAMMAM OMER

Medical AI Researcher & LLM Safety Evaluation Engineer

Omdurman, Sudan | GitHub: github.com/HammamNM | Live Explorer: hammamnm.github.io/reasoning-debugger | Blog: nexoramed.online

Executive Profile: Medical AI researcher and Python framework architect specializing in Clinical LLM Safety Auditing, Neuro-Symbolic Verification, and Automated Rubric Engineering. Creator of the Deterministic-First Hybrid Clinical Auditor, exposing systematic surface-fluency blindness in standalone LLM-as-a-Judge pipelines across 420 emergency clinical evaluations. Bridges clinical training with test-driven software engineering (TDD) and zero-leakage benchmark design.

71.7% (38/53)	0 / 1,767	100% / 100%	< 2 ms CPU
JUDGE BLINDNESS EXPOSED	PARSER HALLUCINATIONS	CRITERION RECALL & SPEC.	STAGE II AUDIT LATENCY

RESEARCH PROJECTS & CORE ENGINEERING DELIVERABLES

1. Deterministic-First Hybrid Clinical Auditor (v0.2-frozen-rules) [GitHub Repo] | [Interactive 420-Trace Explorer] | [50-Scenario Specs]

Lead Architect & Sole Author | Python, Neuro-Symbolic Extraction, Clinical Safety Verification

- **Two-Stage Decoupled Architecture:** Engineered a neuro-symbolic evaluation framework isolating constrained semantic extraction (Stage I) from pure-Python deterministic clinical rule verification (Stage II) across 420 paired clinical management plans on 6 models (openai/gpt-oss-120b, gemini-3.5/3.1-flash-lite, gemma-4-26b-a4b-it, qwen/qwen3.8-27b, openai/gpt-oss-20b).
- **Exposed LLM-as-a-Judge Blindness:** Demonstrated empirically that standalone LLM judges missed 38 of 53 criterion-unsafe responses (71.7% false-pass rate; 95% Wilson CI: 58.4%-82.0%), including 100% of failures (27/27) produced by fluent flagship/large generators under emergency pressure.
- **Resolved Lexical False-Alarm Collapse:** Eliminated the 48.2% false-alarm rate (177/367) of untyped keyword matchers via a canonical Assertion/Negation Gate (PROPOSED vs. NEGATED), rescuing safe clinicians who explicitly document contraindicated drugs to withhold them (100.0% specificity, Youden's J = 1.000).
- **Zero-Hallucination Substring Gate:** Built a deterministic Verbatim Substring Verification Gate enforcing character-level grounding (norm(source_span) in norm(R)), achieving 0 hallucinated extractions across 1,767 verified clinical events.
- **Guideline-Grounded Benchmark:** Authored 50 emergency medicine scenarios (15 dev, 35 strictly held-out test) anchored to Class I recommendations from AHA/ACC, ESC, BTS, EAST, and Surviving Sepsis Campaign guidelines.

2. Reasoning Debugger & Domain-Independent Execution Kernel [Live Cloud App] | [Architecture Spec] | [Kernel Source]

Creator & System Architect | Python, Directed Graph Modeling, Automated Contract Testing

- **Structured Reasoning Graph Engine:** Built a modular Python kernel transforming unstructured chain-of-thought and clinical decision traces into directed acyclic reasoning graphs with explicit hypothesis-evidence provenance links.
- **Cognitive Bias & Failure Detectors:** Implemented deterministic graph analyzers to localize epistemic failure patterns including anchoring bias, premature diagnostic closure, unsupported inference, and temporal prerequisite violations.
- **Production-Grade Contract Testing:** Enforced strict API boundary isolation, serialization round-trip guarantees, and zero-leakage contracts across 30+ automated pytest contract suites.

TECHNICAL & CLINICAL ARSENAL

- **AI Safety & LLM Evaluation:** Multi-Ecosystem Benchmarking, LLM-as-a-Judge Failure Auditing, Sycophancy & Behavioral Shift Analysis, Rubric & Protocol Engineering, Statistical Verification (Wilson 95% CIs, Exact McNemar Tests, Ablation Design).
- **Software Engineering:** Python (OOP, Type-Safe Data Models, CLI & REST APIs), Neuro-Symbolic Pipelines, Deterministic Substring & Regex Engines, JSON Schema Constrained Decoding, Git/GitHub, Linux/Termux CLI Workflows.
- **Clinical Domain Expertise:** Emergency Medicine Triage & Resuscitation Protocols, AHA/ACC, ESC, BTS & ATLS Guideline Interpretation, Differential Diagnosis Modeling, Contraindication & Temporal Prerequisite Auditing.

RESEARCH PUBLICATIONS & PREPRINTS

Omer, H. (2026). *Beyond Surface Fluency: A Deterministic-First Hybrid Clinical Auditor Exposes LLM-as-a-Judge Blindness and Behavioral Shifts Under Emergency Clinical Pressure.* Zenodo Preprint (DOI: 10.5281/zenodo.23835945). Code & Interactive Explorer: github.com/HammamNM/reasoning-debugger

EDUCATION & SCIENTIFIC COMMUNICATION

- **MBBS Candidate (3rd Year):** Faculty of Medicine, Omdurman Islamic University - Omdurman, Sudan (Ongoing, 2026).
- **Founder & Technical Author - NexoraMed (nexoramed.online):** Publishing technical analyses in English on Medical AI evaluation, clinical reasoning frameworks, and AI tools in medical education (2026 - Present).